

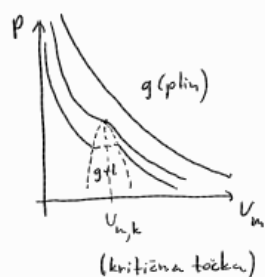
Teorija korespondenčnih stanja

$$\left(p + \frac{a}{V_m^2}\right)(V_m - b) = RT \quad \text{Van der Waals} \quad pV_m = RT \quad \text{id. plin}$$

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$\frac{dp}{dV_m} = -\frac{RT}{(V_m - b)^2} + \frac{2a}{V_m^3}$$

$$\frac{d^2p}{dV_m^2} = \frac{2RT}{(V_m - b)^3} - \frac{6a}{V_m^4}$$



kritična točka:

$$\left. \begin{aligned} p' = 0 & \quad \frac{RT_k}{(V_{m,k} - b)^2} = \frac{2a}{V_{m,k}^3} \\ p'' = 0 & \quad \frac{2RT_k}{(V_{m,k} - b)^3} = \frac{6a}{V_{m,k}^4} \end{aligned} \right\} : \quad \frac{V_{m,k} - b}{2} = \frac{V_{m,k}}{3} \rightarrow \boxed{V_{m,k} = 3b} \rightarrow \frac{RT_k}{(3b - b)^2} = \frac{2a}{(3b)^3} \rightarrow \boxed{T_k = \frac{8}{27} \frac{a}{Rb}}$$

$$\rightarrow p_k = \frac{RT_k}{V_{m,k} - b} - \frac{a}{V_{m,k}^2} = \frac{R \cdot \frac{8}{27} \frac{a}{Rb}}{3b - b} - \frac{a}{(3b)^2} = \frac{a}{27b^2} \left[\frac{8}{3} - 1 \right] = \boxed{\frac{a}{27b^2}}$$

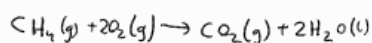
Reducirane količine

$$T_r = \frac{T}{T_k}, \quad p_r = \frac{p}{p_k}, \quad V_r = \frac{V_m}{V_{m,k}}$$

$$p_r p_k = \frac{RT_r T_k}{V_{m,k} V_r - b} - \frac{a}{(V_{m,k} V_r)^2} \rightarrow p_r \frac{R}{27b^2} = \frac{R T_r \frac{8a}{27bR}}{V_r 3b - b} - \frac{a}{V_r^2 9b^2} \rightarrow \boxed{p_r = \frac{8T_r}{3V_r - 1} - \frac{3}{V_r^2}}$$

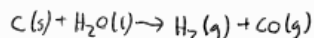
$$Z = \frac{p_k V_{m,k}}{RT_k} = \frac{a}{27b^2} 3b \frac{27bR}{8aR} = \frac{3}{8}$$

$$\left. \begin{aligned} U & \dots \text{notranja energija} & dU &= Tds - pdV \rightarrow \left(\frac{\partial T}{\partial V}\right)_s = -\left(\frac{\partial p}{\partial s}\right)_V \\ H & \dots \text{entalpija} & dH &= Tds + Vdp \rightarrow \left(\frac{\partial T}{\partial p}\right)_s = \left(\frac{\partial V}{\partial s}\right)_p \\ A & \dots \text{prosta energija} & dA &= -SdT - Vdp \rightarrow \left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V \\ G & \dots \text{prosta entalpija} & dG &= -SdT + Vdp \rightarrow \left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p \end{aligned} \right\} \text{Maxwellove relacije}$$



$$T = 25^\circ\text{C}, p = 1\text{bar}, \Delta_r H = -890\text{kJ}$$

eksotermna reakcija, E se sprosti



$$T = 25^\circ\text{C}, p = 1\text{bar}, \Delta_r H = 131\text{kJ}$$

endotermna reakcija, E se porabi

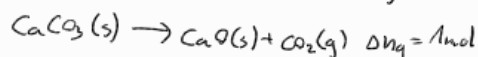
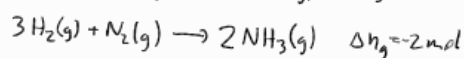
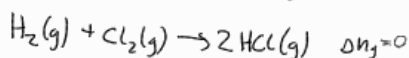
$$\Delta_x, \quad x = \begin{cases} r; \text{reakcija} \\ f; \text{formacija} \\ c; \text{sežig (combustion)} \\ \text{mix; mešanje} \\ \dots \end{cases}$$

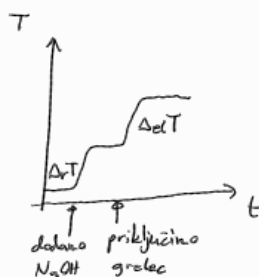
entalpija vs. notranja energija

$$H = U + pV$$

$$\Delta(pV) \approx \Delta n_g RT$$

n snovi v plinastem stanju





$$q_{el} = C \Delta_{el} T = UIt$$

$$q_r = C \Delta_r T$$

$$\rightarrow q_r = q_{el} \frac{\Delta_r T}{\Delta_{el} T}, p = \text{konst.}$$

$$(q_r)_p = \Delta_r H$$

Hessov zakon

- kemijske reakcije so aditivne
- iste računске operacije delamo tudi z $\Delta_r H$

$$\Delta_r H = \sum H(\text{produkti}) - \sum H(\text{reaktanti})$$

Tvorbeni entalpija

- nastane 1 mol spojine iz elementov v referenčnem (najbolj stabilnem) stanju

$$\Delta_{tv} H^\circ \leftarrow \text{oznaka za standardno stanje (p=1 bar, T poljubna)}$$

$$\Delta_{tv}(\text{element v ref. st.}) = 0$$

$$\Delta_{tv}(H_3O^+(aq)) = 0 \text{ dogovor}$$

$$\Delta_r H \text{ [kJ]}$$

$$\Delta_{tv} H \text{ [kJ/mol]}$$

- za večino spojin je $\Delta_{tv} H$ dobljena prek Hessovega zakona

$$\sum \nu_i x_i \rightarrow \sum \nu_j x_j \rightarrow \Delta_r H = \sum_{\text{prod.}} \nu_i \Delta_{tv} H_i^\circ - \sum_{\text{reakt.}} \nu_j \Delta_{tv} H_j^\circ$$

Temp. odv. reakcijske entalpije

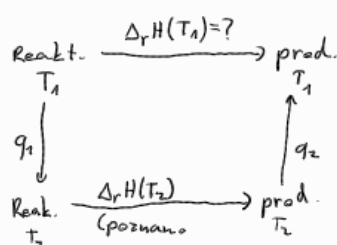
$$\left(\frac{\partial H}{\partial T} \right)_p = C_p$$

$$\Delta_r H^\circ = \sum_{\text{prod.}} \nu_i \Delta_{tv} H_i^\circ - \sum_{\text{reakt.}} \nu_j \Delta_{tv} H_j^\circ \quad \left/ \frac{\partial}{\partial T} \right|_p$$

$$\Delta_r C_p = \sum_{\text{prod.}} \nu_i C_{p,m,i} - \sum_{\text{reakt.}} \nu_j C_{p,m,j}$$

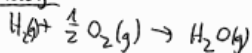
↑ topl. kap. ↑ molarna spec. topl. $\left[\frac{J}{mol \cdot K} \right]$

Kirchoffov zakon



$$\Delta_r H(T_1) = \Delta_r H(T_2) + q_1 + q_2 = \Delta_r H(T_2) + \int_{T_1}^{T_2} \Delta_r C_p dT$$

Naloga



$$\Delta_r H^\circ(H_2O, T=100^\circ C) = ?$$

podatki:

$$\Delta_{tv} H^\circ(H_2O(l), T=25^\circ C) = -241,8 \frac{kJ}{mol}$$

$$C_{p,m}(H_2O(l)) = 28,84 \frac{J}{mol \cdot K}$$

$$C_{p,m}(O_2(g)) = 29,37 \frac{J}{mol \cdot K}$$

$$C_{p,m}(H_2O(g)) = 33,58 \frac{J}{mol \cdot K}$$

$$\begin{aligned} \Delta_r C_p &= \nu(H_2O) C_{p,m}(H_2O) - \left(\nu(H_2) C_{p,m}(H_2) + \nu(O_2) C_{p,m}(O_2) \right) \\ &= 1 \cdot 33,58 \frac{J}{mol \cdot K} - \left(1 \cdot 28,84 \frac{J}{mol \cdot K} + \frac{1}{2} \cdot 29,37 \frac{J}{mol \cdot K} \right) \\ &= -9,945 \frac{J}{K} \end{aligned}$$

$$\begin{aligned} \Delta_r H^\circ(H_2O(g), T=100^\circ C) &= \Delta_r H^\circ(H_2O(l), T=25^\circ C) + \int_{25^\circ C}^{100^\circ C} \Delta_r C_p^\circ dT \\ &= -241,8 kJ - 9,945 \frac{J}{K} \cdot 75 K = \underline{-249,5 kJ} \end{aligned}$$

Sprememba S pri kem. reakc.

$$\Delta_r S^\circ = \sum_{\text{prod}} \nu_i S_{m,i}^\circ - \sum_{\text{reakt.}} \nu_j S_{m,j}^\circ$$

$$S_m^\circ(\text{element}) \neq 0$$

$$S_m(T) = \int_0^{T_{\text{melt}}} \frac{C_{p,m}(l)}{T} dT + \frac{\Delta_{\text{melt}} H^\circ}{T_{\text{melt}}} + \int_{T_{\text{melt}}}^{T_{\text{red}}} \frac{C_{p,m}(l)}{T} dT + \frac{\Delta_{\text{melt}} H^\circ}{T_{\text{melt}}} + \int_{T_{\text{melt}}}^T \frac{C_{p,m}(g)}{T} dT$$

Procesi pri: $V = \text{konst.}$

$$dU = dq + dw = dq - p dV \quad dw = dw_{\text{mech}} (dw_{\text{neut.}}) \quad \text{Procesi pri } p = \text{konst.}$$

$$dU = dq_v \quad dS \geq \frac{dq_v}{T} = \frac{dU}{T}$$

$$\text{če } S = \text{konst.} \quad dU_{v,3} \leq 0$$

Neizvedljivo

$$TdS \geq dU$$

$$T = \text{konst.}$$

$$d(TS) = SdT + TdS$$

$$d(TS) \geq dU$$

$$d(U-TS) \leq 0$$

$$\underbrace{U-TS}_{A} \text{ - prost. energija}$$

$$dH = dq + Vdp$$

$$dH = dq_p$$

$$dS \geq \frac{dq_p}{T}$$

$$TdS \geq dH$$

$$dH_{p,3} \leq 0$$

(neizvedljivo)

$$T = \text{konst.}$$

$$d(H-TS)_{T,p} \leq 0$$

G. prost. entalpija

(Gibbsova prost. en.)

$$\Delta_{\text{tr}} G^\circ(\text{element}) = 0$$

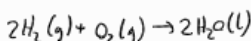
$$\Delta_{\text{tr}} G^\circ(\text{H}^+(\text{aq})) = 0 \quad \text{Dogovor}$$

$$G = H - TS = U + pV - TS$$

$$dG = dU + d(pV) - d(TS) = TdS - p dV + dw_{\text{neut.}} + p dV + V dp - TdS - SdT = -SdT + Vdp + dw_{\text{neut.}}$$

$$(dG)_{T,p} \leq dw_{\text{neut.}}$$

↑
pot. električno



$$\Delta_r H^\circ = 2 \text{ mol} \cdot \Delta_{\text{tr}} H^\circ(\text{H}_2\text{O}(\text{l})) = -2 \cdot 285,8 \text{ kJ}$$

$$S_m^\circ(\text{H}_2(\text{g})) = 130,7 \frac{\text{J}}{\text{mol K}}$$

$$S_m^\circ(\text{O}_2(\text{g})) = 205,2 \frac{\text{J}}{\text{mol K}}$$

$$S_m^\circ(\text{H}_2\text{O}(\text{l})) = 70 \frac{\text{J}}{\text{mol K}}$$

$$\Delta_r S^\circ = \nu(\text{H}_2\text{O}) S_m^\circ(\text{H}_2\text{O}(\text{l})) - (\nu(\text{H}_2) S_m^\circ(\text{H}_2(\text{g})) + \nu(\text{O}_2) S_m^\circ(\text{O}_2(\text{g}))) = -326,6 \frac{\text{J}}{\text{K}}$$

$$G = H - TS$$

$$\Delta_r G^\circ = \Delta_r H^\circ - T \Delta_r S^\circ = -474,2 \text{ kJ}$$

$$\Delta_{\text{tr}} G^\circ = -237,1 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta_r S^\circ = ?$$

$$\Delta_r G^\circ = ?$$

Temp. odv. termodinamskih potencialov

$$\left(\frac{\partial U}{\partial T}\right)_V = C_V$$

$$G = H - TS \quad / : T$$

$$A = U - TS$$

$$\left(\frac{\partial H}{\partial T}\right)_p = C_p$$

$$\frac{G}{T} = \frac{H}{T} - S \quad / \frac{\partial}{\partial T} \Big|_p$$

$$\frac{A}{T} = \frac{U}{T} - S \quad / \left(\frac{\partial}{\partial T}\right)_V$$

$$\left(\frac{\partial S}{\partial T}\right)_p = \frac{C_p}{T}$$

$$\frac{\partial}{\partial T} \left(\frac{G}{T} \right) = \frac{\partial}{\partial T} \left(\frac{H}{T} \right) - \left(\frac{\partial S}{\partial T} \right)_p = \frac{1}{T} \left(\frac{\partial H}{\partial T} \right)_p - \frac{H}{T^2} - \frac{C_p}{T}$$

$$\left(\frac{\partial A}{\partial T}\right)_V = \frac{1}{T} \left(\frac{\partial U}{\partial T}\right)_V - \frac{U}{T^2} - \left(\frac{\partial S}{\partial T}\right)_V$$

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{C_V}{T}$$

$$\left(\frac{\partial G}{\partial T}\right)_p = -\frac{H}{T^2} \quad \text{Gibbs-Helmholtzova enačba}$$

$$\left(\frac{\partial A}{\partial T}\right)_V = -\frac{U}{T^2}$$

Zgled: ΔG za zmrzavanje vode pri $T = -20^\circ\text{C}$

$$\Delta_{\text{melt}} H = 6008 \frac{\text{J}}{\text{mol}} \rightarrow \Delta_{\text{froz}} H = -6008 \frac{\text{J}}{\text{mol}}$$

$$\Delta_{\text{melt}} G(T=0^\circ\text{C}) = 0 \frac{\text{J}}{\text{mol}}$$

$$\frac{\partial(\Delta G/T)}{\partial T} = -\frac{\Delta H}{T^2}$$

$$\int_{T=0^\circ\text{C}}^{T=-20^\circ\text{C}} d\left(\frac{\Delta G}{T}\right) = -\Delta_{\text{froz}} H \int_{T=0^\circ\text{C}}^{T=-20^\circ\text{C}} \frac{dT}{T^2} \rightarrow \left.\frac{\Delta G}{T}\right|_{T=-20^\circ\text{C}} = \left.\frac{\Delta G}{T}\right|_{T=0^\circ\text{C}} - \Delta_{\text{froz}} H \left.\frac{1}{T}\right|_{0^\circ\text{C}}^{-20^\circ\text{C}} = -6008 \frac{\text{J}}{\text{mol}} \left(\frac{1}{253,15\text{K}} - \frac{1}{273,15\text{K}} \right) = -1,738 \frac{\text{J}}{\text{mol K}}$$

$$\Delta_{\text{froz}} G(T=-20^\circ\text{C}) = 253,15\text{K} \cdot (-1,738) \frac{\text{J}}{\text{mol K}} = -440 \frac{\text{J}}{\text{mol}} < 0 \Rightarrow \text{proces je spontan}$$

$p = \text{konst.}$

$$(H_{\text{sistem}})_p = q_{\text{sistem}} = -q_{\text{okolica}}$$

$$\frac{G_{\text{sistem}}}{T} = \frac{H_{\text{sistem}}}{T} - S_{\text{sistem}} = -\frac{q_{\text{okolica}}}{T} - S_{\text{sistem}} = -S$$

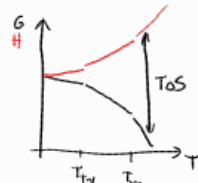
Odlivnost ΔG od T pri $p = \text{konst.}$

$$dG = -SdT + Vdp$$

$$\left(\frac{\partial G}{\partial T}\right)_p = -S < 0$$

$$S(s) < S(l) < S(g)$$

$$\left(\frac{\partial H}{\partial T}\right)_p = c_p > 0$$



Odlivnost ΔG od p pri $T = \text{konst.}$

$$dG = -SdT + Vdp$$

a) kondenzirani snovi (s, l)

$$V \approx \text{konst.}$$

$$dG_n = V dp$$

$$G_n(p_2) = G_n(p_1) + V_n(p_2 - p_1)$$

$$p_1 = 1 \text{ bar} = p^*, p_2 = p$$

$$G_n(p) = G_n^* + V_n(p - p^*)$$

b) plini (idealni) $pV_n = RT$

$$dG_n = V_n dp = RT \frac{dp}{p}$$

$$G_n(p_2) = G_n(p_1) + RT \ln \frac{p_2}{p_1}$$

$$p_2 = p^{id}, p_1 = 1 \text{ bar} = p^*$$

$$G_n(p^{id}) = G_n^* + RT \ln \frac{p}{p^*}$$

c) plini (realni)

$$z = \frac{pV_n}{RT} = 1 + B_2(T)p + B_3(T)p^2 + \dots$$

virialna enačba stanja

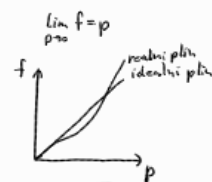
$$V_n = RT \left(\frac{1}{p} + B_2 + B_3 p + \dots \right)$$

$$G_n(p_2) = G_n(p_1) + RT \left(\ln \frac{p_2}{p_1} + B_2(p_2 - p_1) + \frac{B_3}{2}(p_2^2 - p_1^2) + \dots \right)$$

$$p_2 = p, p_1 = p^{id}$$

$$G_n(p) = G_n^* + RT \left(\ln \frac{p}{p^*} + B_2(p - p^{id}) + \frac{B_3}{2}(p^2 - p^{id2}) + \dots \right)$$

Lewis: f -fugativnost (konkacija Haka)



$$\rightarrow G_n(p) = G_n(p^*) + RT \ln \frac{f}{p^*}$$

standardna stanja za plin:

$$p = p^* = 1 \text{ bar}$$

idealno obnašanje

Odlivnost f od p pri $T = \text{konst.}$

$$dG_n = RT d \ln f$$

$$dG_n = V_n dp$$

$$RT d \ln f = V_n dp \quad / : RT$$

$$d \ln f = \frac{V_n}{RT} dp \quad / - d \ln p$$

$$d \ln \frac{f}{p} = \frac{V_n}{RT} dp - d \ln p = \left(\frac{V_n}{RT} - \frac{1}{p} \right) dp \quad / \int$$

$$\ln \frac{f}{p} - \ln \frac{f^*}{p^*} = \int_{p^*}^p \left(\frac{V_n}{RT} - \frac{1}{p} \right) dp$$

primer: O_2 , $p = 1 \text{ bar}$, poenostavljeni vdW enačba $p(V_n - b) = RT$, $b = 2,11 \cdot 10^{-5} \frac{m^3}{mol}$

1) $T = 25^\circ C$, $p = 1 \text{ bar}$, $f = ?$

$$\ln \frac{f}{p} = \int_{p^*}^p \left(\frac{RT}{p^2(b + \frac{RT}{p})} - \frac{1}{p} \right) dp = \frac{b}{RT} p = 8,51 \cdot 10^{-4}$$

$$f = p e^{8,51 \cdot 10^{-4}} = p \cdot 1,000851$$

Aktivnost

$$a = \frac{f}{f^*}$$

Komponenta vs. sestavina

- voda(l) 1komp., 1sest.

- voda(g) 1komp., 1sest.

- voda + etanol(l) 2komp., 1sest.

- voda + NaCl(aq) $\rightarrow H_2O + Na^+ + Cl^-$

2komp., 3sest., $n(Na^+) = n(Cl^-)$

#komp. $\approx$ #sest. - #matematičnih poverjav

- $PCl_5 \rightarrow PCl_3 + Cl_2$ $n(PCl_5) = n(Cl_2)$

Kemijski potencial

$$dG = -SdT + Vdp + \sum_i \mu_i dn_i$$

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T, p, n_{j \neq i}} \quad \text{za kak se spremeni } G, \text{ če dodamo 1 mol snovi v sistem}$$

$$dU = TdS - pdV + \sum_i \left(\frac{\partial U}{\partial n_i} \right)_{S, V, n_{j \neq i}} dn_i$$

$$dH = TdS + Vdp + \sum_i \left(\frac{\partial H}{\partial n_i} \right)_{p, S, n_{j \neq i}} dn_i$$

$$dA = -SdT - pdV + \sum_i \left(\frac{\partial A}{\partial n_i} \right)_{T, V, n_{j \neq i}} dn_i$$

$$\left(\frac{\partial U}{\partial n_i} \right) = \left(\frac{\partial H}{\partial n_i} \right) = \left(\frac{\partial A}{\partial n_i} \right) = \left(\frac{\partial G}{\partial n_i} \right)$$

Večkompontentni vezfazi sistem

$$dG = \sum_{j=1}^p dG^j \quad p \dots \text{število faz}$$

$$dG = - \sum_{j=1}^p S^j dT + \sum_{j=1}^p V^j dp + \sum_{j=1}^p \sum_{i=1}^c n_i^j d\mu_i^j$$

$$T, p = \text{konst.}, \quad c=1, p=2$$



dG_{SO}

$$\mu^g d n^g + \mu^l d n^l \leq 0$$

$$\mu^g + n^g = \text{konst.}$$

$$d\mu^g + d\mu^l = 0$$

$$d\mu^l = -d\mu^g$$

$$\mu^g d n^g \leq \mu^l d n^g$$

izparevanje ravnotežje

$$d\mu^g < 0$$

$$\mu^g = \mu^l$$

$$\mu^g > \mu^l$$

$$\mu_1^g = \mu_1^l = \dots = \mu_1^p \quad \text{komponenta 1}$$

$$\mu_c^g = \mu_c^l = \dots = \mu_c^p \quad \text{komponenta c}$$

(p-1)c enačb

$$x_i = \frac{n_i}{n} \quad \text{molski delež}$$

$$\sum_{i=1}^c x_i^g = 1, \quad \sum_{i=1}^c x_i^l = 1, \dots, \quad \sum_{i=1}^c x_i^p = 1$$

(c-1)p + 2 neznanke

$$\downarrow$$

$$T, p$$

$$f = (c-1)p + 2 - (p-1)c = c - p + 2$$

$$f = c - p + 2 \quad \text{Gibbsovo fazno pravilo}$$

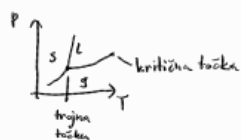
število prost. stopenj

Primer: trojna točka

$$c=1, p=3$$

$$f=0$$

→ vse neznanke so točno določene



Okoli trojne točke

$$\Delta_{\text{sub}} H = \Delta_{\text{td}} H + \Delta_{\text{vap}} H$$

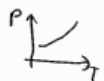
okoli kritične točke

$$\Delta H = T \Delta S = T(S_g - S_l) = 0$$

$$g-l \quad c=1, p=2$$

$$f=1$$

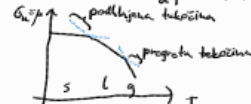
1 izmed neznanek lahko izberemo (npr. če izberemo T , lahko določimo p)



$$dG = -S dT + V dp \quad / \quad \frac{1}{n} \quad \text{čista snov}$$

$$dG_m = \mu = -S_m dT + V_m dp \quad \mu = \frac{G}{n}$$

$$p = \text{konst.} \quad S_m = \frac{dG_m}{dT} \neq f(T)$$



$$\mu^l(T, p) = \mu^g(T, p) \quad / \quad d$$

$$\left(\frac{\partial \mu^l}{\partial T} \right)_p dT + \left(\frac{\partial \mu^l}{\partial p} \right)_T dp = \left(\frac{\partial \mu^g}{\partial T} \right)_p dT + \left(\frac{\partial \mu^g}{\partial p} \right)_T dp$$

$$-S_m^l dT + V_m^l dp = -S_m^g dT + V_m^g dp$$

$$(S_m^g - S_m^l) dT = (V_m^g - V_m^l) dp$$

$$\frac{dp}{dT} = \frac{S_m^g - S_m^l}{V_m^g - V_m^l} = \frac{\Delta_{\text{vap}} S}{V_m^g - V_m^l}$$

Clapeyron

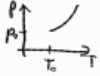
$$\frac{\Delta_{\text{vap}} H}{T} = \Delta_{\text{vap}} S$$

$$V_h^l \ll V_h^g = \frac{RT}{p} :$$

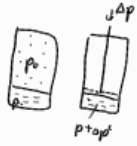
$$\frac{dp}{dT} = \frac{\Delta_{izp} H \cdot p}{RT^2}$$

$$\int_{p_0}^p \frac{dp}{p} = \int_{T_0}^T \frac{\Delta_{izp} H}{RT^2} dT, \text{ predp. } \Delta_{izp} H \neq f(T)$$

$$\ln \frac{p}{p_0} = \frac{\Delta_{izp} H}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right)$$

$$T > T_0 \Rightarrow p > p_0$$


Odvishnost parnega tlaka od zunanjskega tlaka

$$\mu^g(T, p^g) = \mu^l(T, p^l)$$


$$\left(\frac{\partial \mu^g}{\partial p^l} \right)_T = \left(\frac{\partial \mu^l}{\partial p^l} \right)_T$$

$$\left(\frac{\partial \mu^g}{\partial p^g} \right)_T \frac{\partial p^g}{\partial p^l} = \left(\frac{\partial \mu^l}{\partial p^l} \right)_T$$

$$V_h^g \frac{dp^g}{dp^l} = V_h^l$$

$$dp^g = \frac{V_h^l}{V_h^g} dp^l, \quad 0 < \frac{V_h^l}{V_h^g} \ll 1$$

$$\int_{p_0}^p \frac{dp^g}{p^g} = \frac{V_h^l}{RT} \int_{p_0}^{p+dp^l} dp^l$$

$$\ln \frac{p}{p_0} = \frac{V_h^l}{RT} (p+dp^l - p_0) \approx \frac{V_h^l}{RT} dp^l$$